MILITARY SPECIFICATION

PREFIRE TEST ASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of test assembly entitled Prefire Test Assembly which is used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8825243 - Prefire Test Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

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3.2 Assembly. The prefire test assembly shall comply with all requirements specified on the drawing (dwg) 8825243, with all requirements specified to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Performance.

3.4.1 Starting torque. The torque required to turn the starting generator shaft shall not exceed 13.5 inch pounds maximum when tested as specified in 4.4.1.1.

3.4.2 Continuity. The assembly shall exhibit continuity as indicated by applicable drawings and shall be capable of satisfactory operation when tested as specified in 4.4.1.2.

3.5 Environmental testing. The assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The assembly shall also comply with 3.4 before and after each environmental test. Testing shall be as specified in 4.4.1. Tests shall be conducted in the order specified.

3.5.1 Vibration. The assembly shall be capable of withstanding vibration in any direction without evidence of damage such as loose parts and shall comply with the requirments of 3.4 after being tested as specified in 4.4.2.1.

3.5.2 Humidity. The assembly shall show no evidence of electrical or mechanical damage such as visible loosening or deformation of any part when tested as specified in 4.4.2.2.

3.6 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Burrs. Parts shall be free from burrs.

3.7.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.7.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8825243) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Performance (see 3.4)		X
3. Vibration (see 3.5.1)		X
4. Humidity (see 3.5.2)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one shift, one batch of products, one manufacturer or raw material, one manufacturer of components, one production method, and one manufacturing operation. The entire lot shall be submitted as a unit for inspection.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

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4.3.2.1.1 Prefire test assembly prior to closing terminal board and toggle switch (see dwg 8825243).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper length of wire	Gage
102.	Total length, min	Gage
103.	Cold solder connection (see 3.6)	Visual
104.	Improper color code	Visual
105.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.1.2 Prefire test assembly (see dwg 8825243).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Insulation torn or damaged	Visual
102.	Lacing broken or loose	Visual/Manual
103.	Hardware loose or missing	Visual/Manual
104.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Marking incorrect or illegible	Visual
202.	Evidence of poor workmanship (see 3.7)	Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation packing and packaging not in accordance with MIL-STD-1169	Visual

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102. Marking not in accordance with MIL-STD-1169.....Visual

Minor:

201. Evidence of poor workmanship (see 3.7).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test classification
a.	Performance	Critical
b.	Vibration	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Performance. Each prefire assembly in the lot shall be tested in accordance with 4.4.1. If any item fails to comply with 3.4, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration. One sample prefire assembly shall be selected from every 25 prefire assemblies or portion thereof produced. The sample shall be tested in accordance with 4.4.2.1. If any item fails to comply with 3.5.1 the lot shall be refected. The first articles test sample will satisfy the first lot sampling requirement.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions designed or specified by the contractor. The contractor shall obtain approval of such designs, operating instructions and calibration instructions prior to use on the contract from Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I as specified on DI-R-1714 (see 6.1.2).

4.3.4.2 Minor characteristics and tests. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Performance.

4.4.1.1 Starting torque. The torque of the starting generator shaft shall be measured with a torque wrench.

4.4.1.2 Continuity. When operated in the counter-clockwise direction at 70 rpm both the red and green light shall light. When the generator is operated in the clock-wise direction at 70 rpm with the toggle switch placed in GO position (Same side as positioning key) the green light shall light. When the generator is operated in the clockwise direction at 70 rpm with the toggle switch placed in NO-GO position (Side opposite to positioning key) the red light shall light.

4.4.2 Environmental testing (see 3.5).

4.4.2.1 Vibration. The terminal board assembly, toggle switch, light bracket and generator shall be rigidly mounted to the vibration table. A simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion) shall be applied, the frequency being varied uniformly between the approximate limits of 10 and 55 Hertz (Hz). The entire frequency range from 10 to 55 to 10 Hz shall be transversed in approximately one minute. The motion shall be continuously applied for 15 minutes in each of the three mutually perpendicular directions. After completion of this test the assembly shall be tested as specified in 4.4.1.

4.4.2.2 Humidity. The assembly shall be loosely coiled and placed in a humidity test chamber. Within a period of one hour, the test chamber temperature shall be raised to 165 ± 5 degrees Fahrenheit (°F) and held for a period of two hours. A relative humidity of 95 percent minimum, as measured by a psychrometric instrument, shall be maintained while the chamber temperature is 165 ± 5°F. At the end of the two hour period the chamber temperature shall be reduced uniformly over a one hour-period to between 68° and 95°F. During the period of decreasing temperature, the relative humidity shall be 100 percent. The assembly shall be subjected to a total of ten humidity cycles (one hour increasing temperature, two hours of temperature stabilization (see 6.4) and 95 percent minimum relative humidity, and one hour decreasing temperature at 100 percent relative humidity). Distilled water having a pH value of between 6.5 and 7.5 at 77°F shall be used in controlling the relative humidity. The pH value shall be measured electrometrically using a glass electrode with a saturated potassium chloridde bridge, or other method of equivalent accuracy. Only chemically pure hydrochloric acid or chemically pure sodium hydroxide shall be used to adjust the pH value.

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After completion of the tenth cycle the assembly shall be removed from the test chamber immediately and dried by blowing dry, compressed, filtered air over the assemblies for a period of five minutes. After completion of this test the assembly shall be tested as specified in 4.4.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The prefire assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The prefire assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The prefire assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1815 dated 26 February 1962.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A145

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